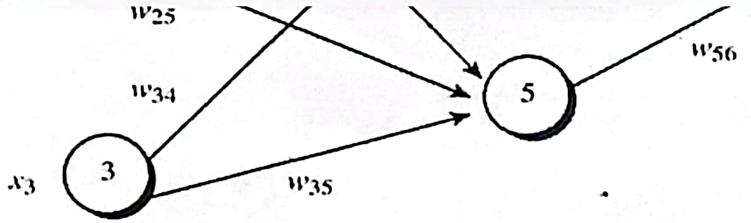

x1	x2	x3	w14	w15	w24	w25	w34	w35	w46	w56	b4	b5	b6
1	0	1	-0.2	-0.3	0.1	0.1	0.5	-0.2	-0.3	-0.2	-0.4	0.2	0.1

Use **hyperbolic tangent** on neuron 4 & 5. Use **Sigmoid** on neuron 6.

Neuron 4:

$z_4 = (x_1 . w_{14} + x_2 . w_{24}) + (x_3 . w_{34}) + b_4$

$z_4 = -0.1$

$a_4 = \tanh(-0.1) = -0.0997$

Neuron 5:

$z_5 = (x_1 . w_{15}) + (x_2 . w_{25}) + (x_3 . w_{35}) + b_5$

$z_5 = -0.3$

$a_5 = -0.2913$

Output layer :

$$z_6 = (a_4 . W_{46}) + (a_5 . W_{56}) + b_6$$

$$z_6 = 0.18817$$

$$a_6 = \sigma(z_6) = 0.5469$$

Actual $Y = 1$

$$L = -[y \log(\hat{y}) + (1-y)\log(1-\hat{y})]$$

$$L = 0.6035 \qquad \eta = 0.01$$

BACK PPOR

WEIGHT UPDATES

$$\frac{\partial L}{\partial w_{46}} = \frac{\partial L}{\partial a_6} \cdot \frac{\partial a_6}{\partial z_6} \cdot \frac{\partial z_6}{\partial w_{46}}$$

$$\delta_6 = a_6 - y = 0.5469 - 1 = -0.4531$$

simplified for sig + BCE

$$\boxed{\frac{\partial L}{\partial w_{46}} = \delta_6 \cdot a_4 = (-0.4531)(0.0997) = 0.04517}$$

$$\frac{\partial L}{W_{14}} = \delta_6 \cdot W_{46} \cdot g'(z_4) . x_1$$

$$g'(z_4) = 1 - (a_4)^2 = 1 - (-0.0997)^2 \approx 0.9901$$

$$\delta_4 = (\delta_6 . w_{46}) . g'(z_4) = (-0.4531 - 0.3) . 0.9901$$

$$\approx 0.1346$$

$$\boxed{\frac{\partial L}{W_{14}} = (0.1346) . x_1 = 0.1346}$$

$$W_{46}^{new} = W_{46} - \eta \frac{\partial L}{\partial W_{46}} = -0.3 - (0.01)(0.04517) = -0.3004517$$

$$W_{14}^{new} = W_{14} - \eta \frac{\partial L}{\partial W_{14}} = -0.2 - (0.01)(0.1346) = -0.201346$$